Project 6:-Maven Selenium Project – Firefox Browser

Step 1: Create a Maven Project

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenFirefoxApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
harith@harith-VirtualBox:~$ cd devops
harith@harith-VirtualBox:~/devops$ cd test
harith@harith-VirtualBox:~/devops/test$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenFirefoxApp -DarchetypeArtifactId=maven-ar
chetype-quickstart -DinteractiveMode=false
Project Structure:
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/harith/devops/test
[INFO] Parameter: package, Value: com.example
```

Step 2: Install Firefox Browser

firefox –version

if not installed: sudo apt install firefox

Step 3: Install GeckoDriver

wget https://github.com/mozilla/geckodriver/releases/latest/download/geckodriver-v0.34.0-linux64.tar.gz

tar -xvzf geckodriver-v0.34.0-linux64.tar.gz

sudo mv geckodriver /usr/local/bin/

sudo chmod +x /usr/local/bin/geckodriver

geckodriver --version

Step 4: Add Selenium Dependency in pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/maven-v4_0_0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.example</groupId>
<artifactId>MyMavenFirefoxApp</artifactId>
<version>1.0-SNAPSHOT</version>
<packaging>jar</packaging>


<name>MyMavenFirefoxApp</name>
<dependencies>
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>selenium-java</artifactId>
<version>4.29.0</version>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.11.0</version>
<configuration>
<source>11</source>
<target>11</target>
</configuration>
</plugin>
<plugin>
<groupId>org.codehaus.mojo</groupId>
<artifactId>exec-maven-plugin</artifactId>
```

```
<version>3.1.0</version>
<configuration>
<mainClass>com.example.App</mainClass>
</configuration>
</plugin>
</plugins>
</build>
</project>
```

Step 5: Inspect Elements in the Websites

Steps:

1. Open the website.
2. Right-click on the webpage.
3. Click **Inspect**.
4. Dentify the required HTML elements.

Step 6: Create App.java

Location:

src/main/java/com/example/App.java

Java Code:

```
package com.example;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.WindowType;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;

public class App {

    public static void main(String[] args) throws InterruptedException {

        FirefoxOptions options = new FirefoxOptions();

        WebDriver driver = new FirefoxDriver(options);

        driver.manage().window().setSize(new org.openqa.selenium.Dimension(1920,1080));
```

```
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(15));

Actions actions = new Actions(driver);

driver.get("https://www.saucedemo.com/");

driver.findElement(By.id("user-name")).sendKeys("standard_user");

driver.findElement(By.id("password")).sendKeys("secret_sauce");

driver.findElement(By.id("login-button")).click();

System.out.println("SauceDemo login successful");

driver.switchTo().newWindow(WindowType.TAB);

driver.get("https://automationexercise.com/products");

driver.findElement(By.id("search_product")).sendKeys("Men Tshirt");

driver.findElement(By.id("submit_search")).click();

WebElement product = wait.until(
        ExpectedConditions.visibilityOfElementLocated(
                By.cssSelector("a[data-product-id='2']")
        )
);

((org.openqa.selenium.JavascriptExecutor) driver)
        .executeScript("arguments[0].scrollIntoView({block:'center'});", product);

((org.openqa.selenium.JavascriptExecutor) driver)
        .executeScript("arguments[0].click();", product);

WebElement viewCart = wait.until(
        ExpectedConditions.elementToBeClickable(
                By.cssSelector("#cartModal a[href='/view_cart']")
        )
);

((org.openqa.selenium.JavascriptExecutor) driver)
        .executeScript("arguments[0].scrollIntoView(true);", viewCart);

((org.openqa.selenium.JavascriptExecutor) driver)
        .executeScript("arguments[0].click();", viewCart);

System.out.println("Automation Exercise product added to cart");

driver.switchTo().newWindow(WindowType.TAB);

driver.get("https://practicetestautomation.com/practice-test-login/");
```

```
        Thread.sleep(2000);

        driver.findElement(By.id("username")).sendKeys("student");

        driver.findElement(By.id("password")).sendKeys("Password123");

        driver.findElement(By.id("submit")).click();

        System.out.println("Practice Test Automation login successful");

        Thread.sleep(5000);

        driver.quit();
    }
}
```

Step 7: Build the Maven Project

Compile and build the project using the command:

mvn clean install

```
harith@harith-VirtualBox:~/devops/test/MyMavenFirefoxApp$ mvn clean compile
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenFirefoxApp >--------------------
[INFO] Building MyMavenFirefoxApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenFirefoxApp ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenFirefoxApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/harith/devops/test/MyMavenFirefoxApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ MyMavenFirefoxApp ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 11] to target/classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.788 s
[INFO] Finished at: 2026-03-17T22:34:45+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test/MyMavenFirefoxApp$
```

Step 8: Run the Application

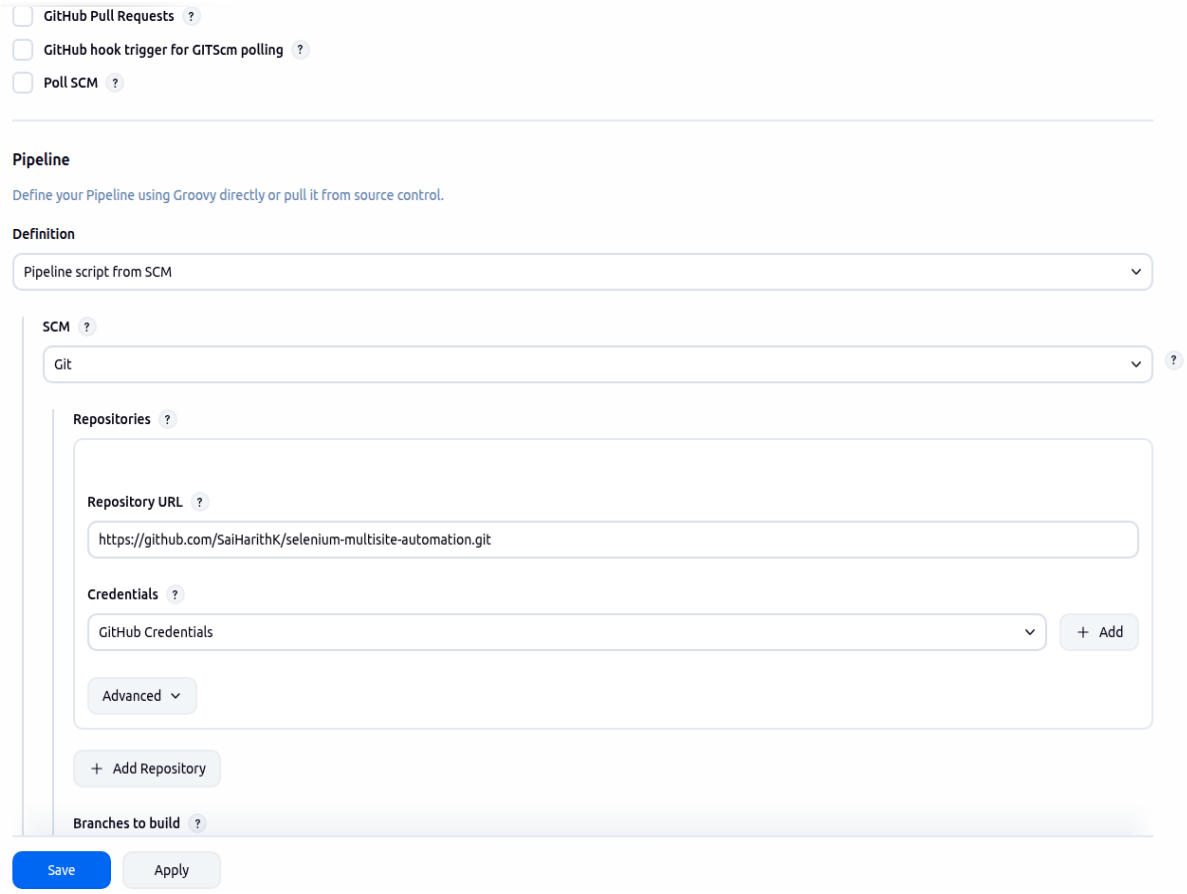
Run the application using: mvn exec:java

```
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test/MyMavenFirefoxApp$ mvn exec:java
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenFirefoxApp >--------------------
[INFO] Building MyMavenFirefoxApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.1.0:java (default-cli) @ MyMavenFirefoxApp ---
```

Step 9: Pipeline

-login to jenkins account

-create a new item and configure the following settings



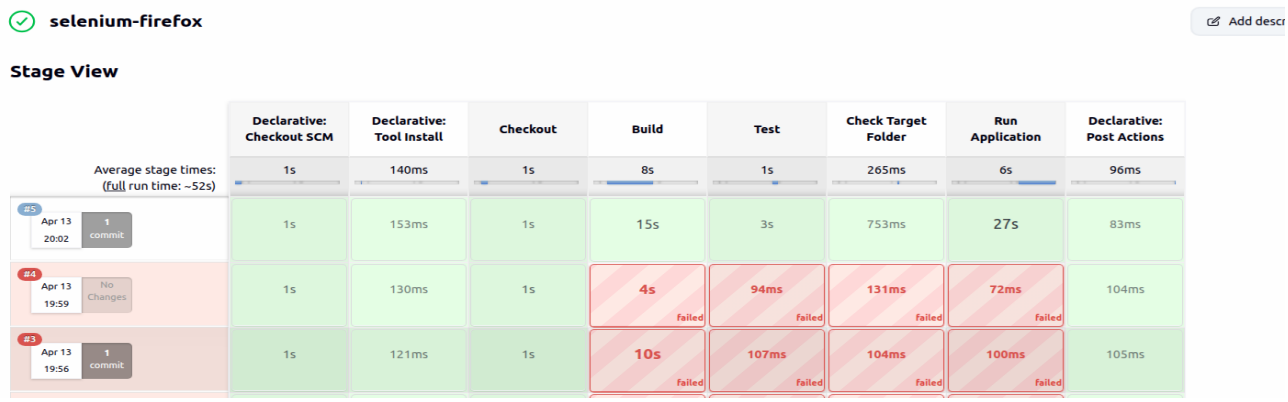
Step 10: Edit Jenkins file

-Create a Jenkinsfile and add the following code:

pipeline {

```
agent any  // Use any available agent
tools {
    maven 'Maven'  // Ensure this matches the name configured in Jenkins
}
stages {
    stage('Checkout') {
        steps {
            git branch: 'main', url: 'https://github.com/SaiHarithK/selenium-multisite-automation.git'
        }
    }
    stage('Build') {
        steps {
            sh 'mvn clean package'  // Run Maven build
        }
    }
    stage('Test') {
        steps {
            sh 'mvn test'  // Run unit tests
        }
    }
    stage('Run Application') {
        steps {
            // Start the JAR application
            sh 'java -jar target/SeleniumMultiSite-1.0-SNAPSHOT.jar'
        }
    }
}
post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
```

```
            echo 'Build failed!'
        }
    }
}
```

Step 11: Build

Click on build now and check if its successfull